

C1_AIB0326 · ISB ALP TEAM 10 · REVIEW 3

iADAS

Intelligent Anomaly Detection & Alerting System

AI-Powered Retail Decision Support using Forecasting & Generative AI.

Business Problem

Problem Statement · Retail managers monitor thousands of daily transactions but often struggle to identify abnormal sales patterns in time.

Business Challenges

Manual anomaly detection

Reactive decision-making

Limited business interpretation

Delayed management reporting

Project Objective

✓ Forecast sales proactively

✓ Detect abnormal business events

✓ Identify likely business drivers

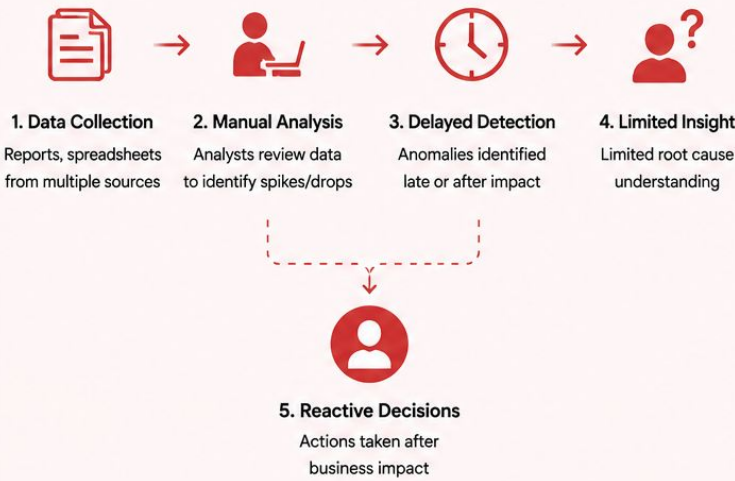
✓ Deliver AI-powered executive recommendations

From Reactive Reporting to Proactive Decisions

CURRENT PROCESS

Manual. Reactive. Time-Consuming.

HOW IT WORKS TODAY



PAIN POINTS

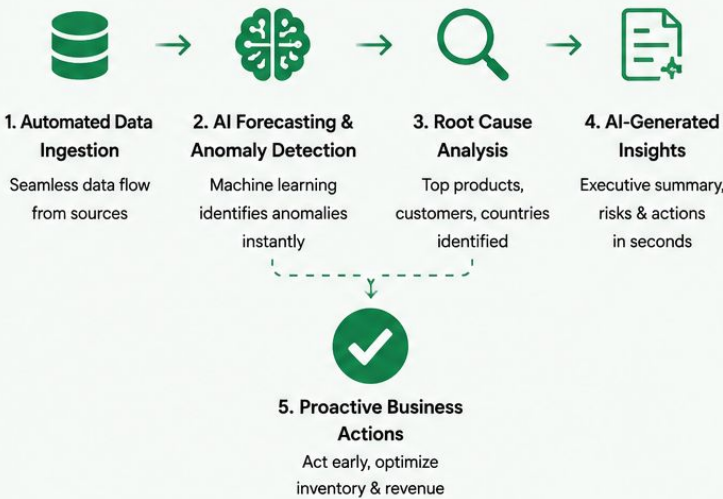
- ✗ Time-consuming manual work
- ✗ Hidden anomalies
- ✗ Missed opportunities
- ✗ Delayed actions
- ✗ Impact on revenue & inventory
- ✗ Higher operational effort

VS

AI-POWERED PROCESS

Automated. Proactive. Intelligent.

HOW IT WORKS WITH iADAS



KEY BENEFITS

- ✓ Automated & faster
- ✓ Early anomaly detection
- ✓ Deep business insights
- ✓ Proactive decision making
- ✓ Improved inventory planning
- ✓ Better business outcomes



Outcome: From reactive reporting to proactive decision intelligence.

Dataset & Data Preparation

Online Retail II

UCI / Kaggle · UK online retailer

Dec 2009 – Dec 2011

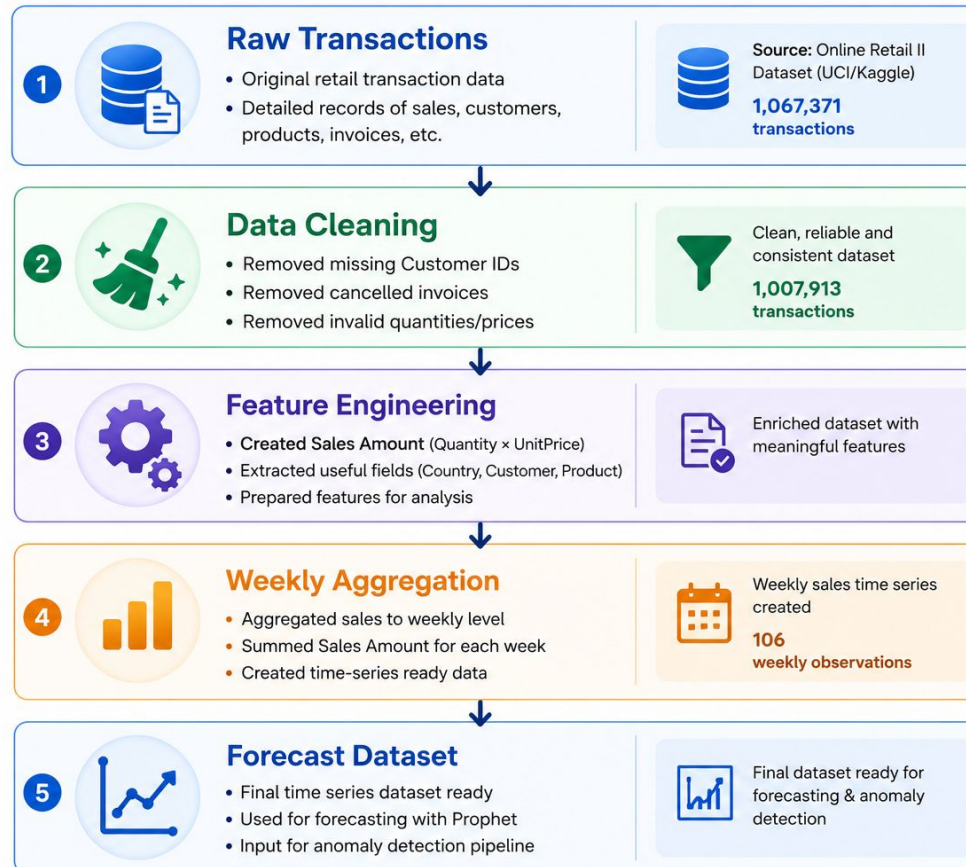
Data collection period

1,067,371

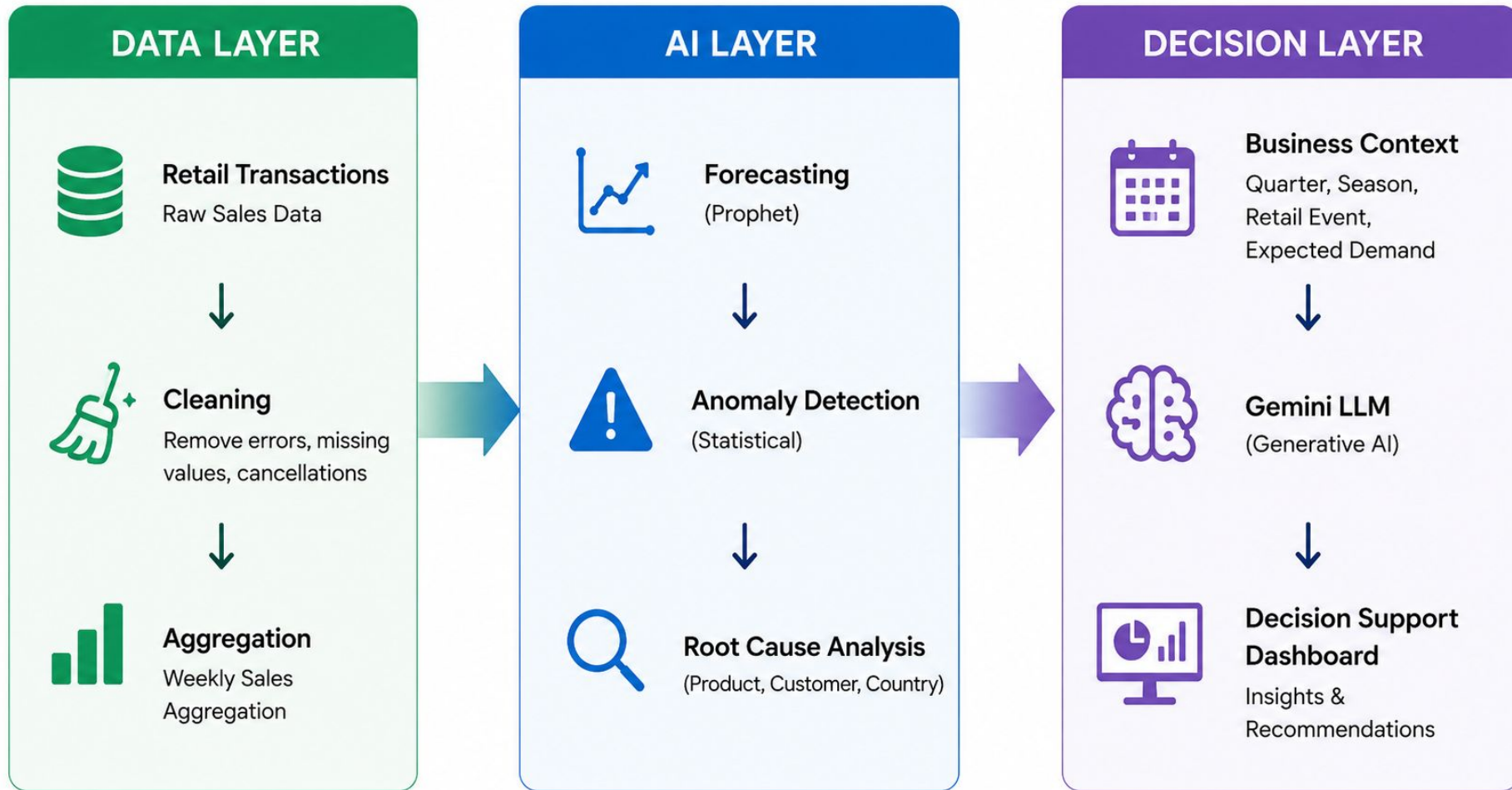
Original transactions

40+ Countries

Thousands of products & customers



Methodology



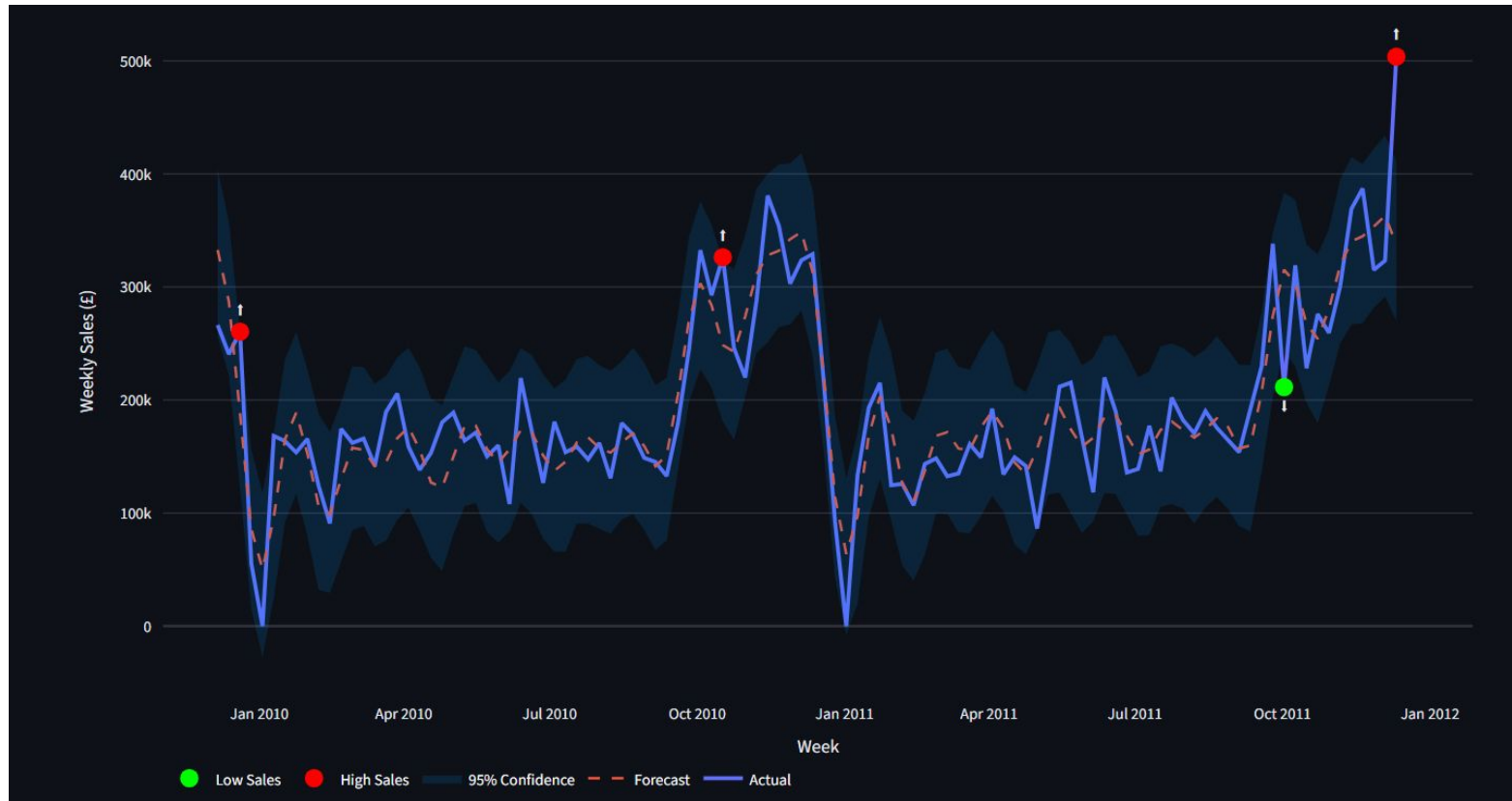
Forecasting & Anomaly Detection

27,549
MAE

36,860
RMSE

£193,172
Avg Weekly Sales

4
Anomalies

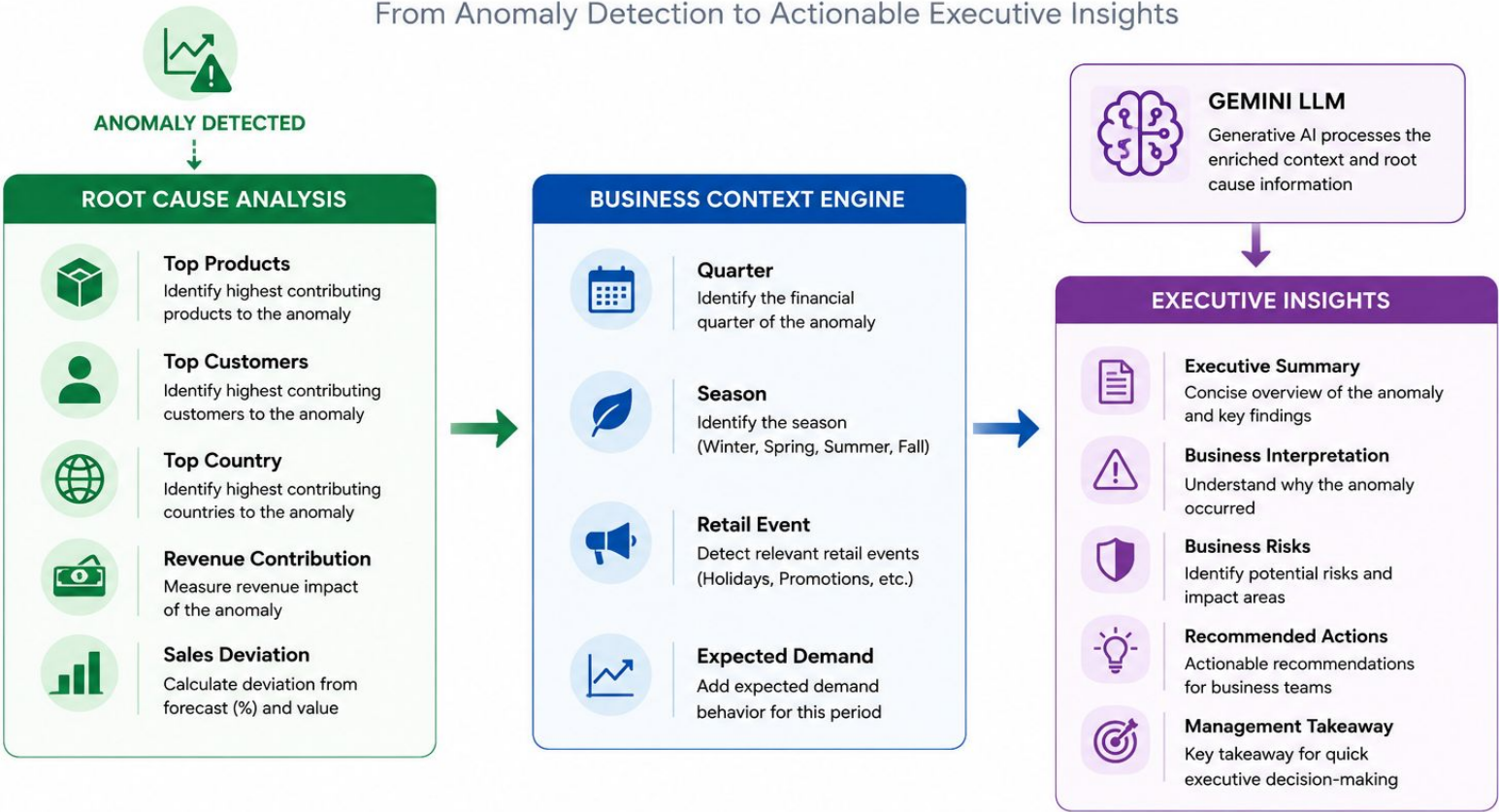


Weekly sales are forecast using Prophet. Observations outside the prediction interval are automatically classified as anomalies.

Root Cause Analysis & Business Context

Root Cause Analysis & Business Context

From Anomaly Detection to Actionable Executive Insights



Combining data-driven analysis with business context and Generative AI delivers accurate, relevant and actionable executive insights.

AI Product Demo

Features

- ✓ Forecast Visualization
- ✓ Interactive Anomaly Selection
- ✓ KPI Dashboard
- ✓ Root Cause Analysis
- ✓ AI Executive Summary
- ✓ Recommended Actions



WHY IT MATTERS

Business Value

TRADITIONAL REPORTING	IADAS
Manual monitoring	Automated
Static dashboards	AI-powered insights
Reactive decisions	Proactive alerts
Manual analysis	Executive summaries
Hours of analysis	Minutes

Expected Business Benefits

 **Faster Detection**

 **Better Visibility**

 **Improved Inventory**

 **Automated Reporting**

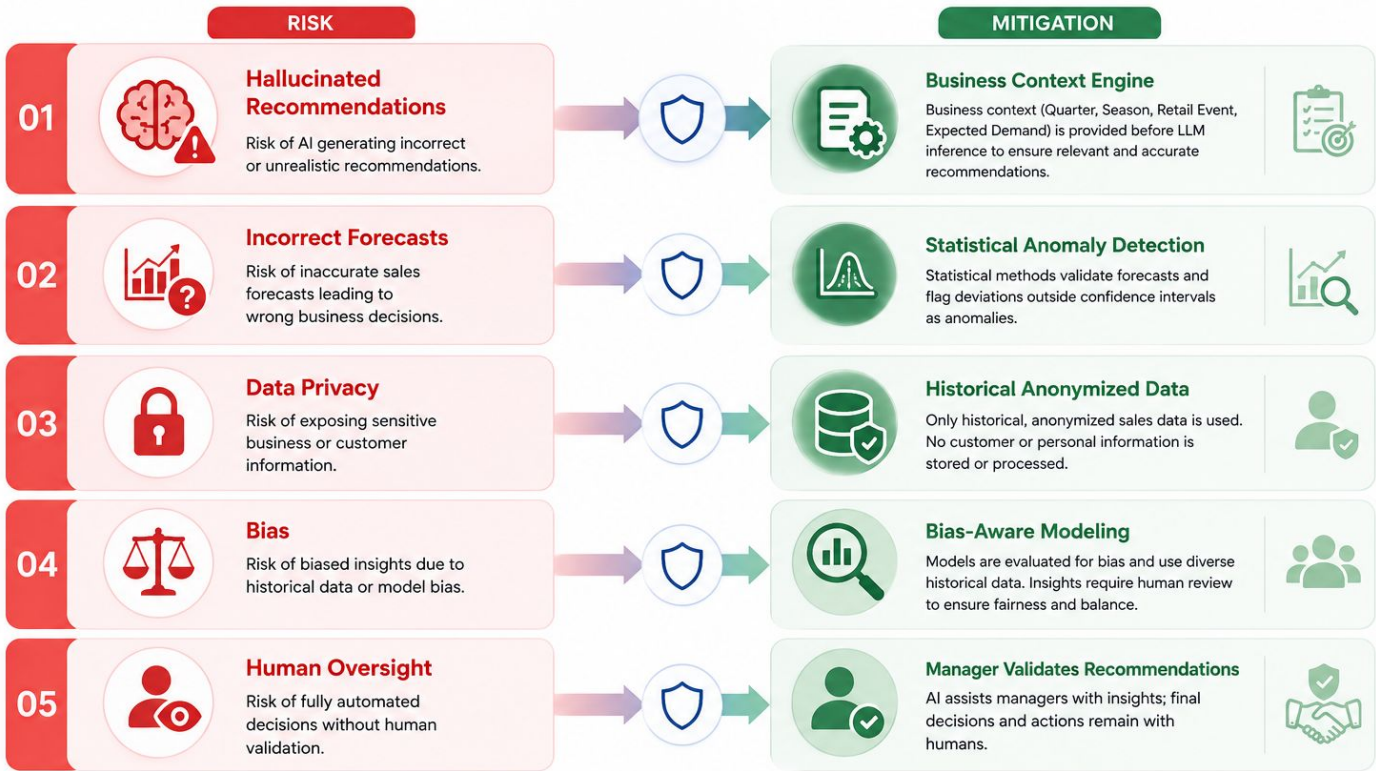
 **Better Decisions**

Responsible AI

AI recommendations are designed to support—not replace—human decision-making.

AI Risks & Mitigations

Responsible AI for Reliable & Actionable Insights



AI ASSISTS. HUMANS DECIDE. BUSINESS IMPACT DELIVERED.

Building trust through transparency, validation & responsibility.

WRAPPING UP

Conclusion

Project Achievements

- ✓ Built an end-to-end AI-powered retail decision support system
- ✓ Forecasted weekly retail sales
- ✓ Automatically detected anomalies
- ✓ Performed root cause analysis
- ✓ Generated executive insights using Gemini
- ✓ Delivered an interactive Streamlit dashboard

iADAS

iADAS demonstrates how predictive analytics combined with Generative AI can transform retail transaction data into actionable executive decision support.